

IoT and AI-Driven Adaptive Traffic Signal Control Using Deep Reinforcement Learning and Graph Neural Networks

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Abstract—The urban transportation system suffers from severe congestion due to rapid growth in vehicle density, limited road infrastructure, and inefficient traffic signal control. Fixed-time and actuated controllers fail to adapt with the dynamics of varying traffic conditions, which eventually results in lengthy delays, higher consumption of fuel, and increased emissions. This work proposes an IoT and AI-driven adaptive traffic signal control framework by incorporating DRL and GNN methodologies for real-time optimization. The continuous V2I communication facilitated by IoT-enabled RSUs and OBUs provides real-time live data related to queue length, vehicle density, and waiting time. The DRL agent incorporates several algorithms, namely, DQN, PPO, and A3C, to calculate the optimal phase duration. Further, GNNs model the spatial dependencies among intersections for enabling coordinated network-level control. The concept of edge computing has also been exploited to reduce latency and increase scalability. Experimental validation has been done in the SUMO environment, where performance metrics like average delay, queue length, and emission level have been analyzed. The proposed IoT-AI-GNN hybrid ensures improved traffic flow, reduced waiting time, and enhanced environmental sustainability, thus proving its viability for smart city applications.

Keywords—*Internet of Things (IoT), Deep Reinforcement Learning (DRL), Graph Neural Networks (GNN), Adaptive Traffic Control, Edge Computing, Smart Cities*

I. INTRODUCTION

Traffic congestion is among the biggest challenges rapidly urbanizing cities are facing. Exponential growth in vehicles, coupled with static traffic management systems, has resulted in a lot of inefficiencies. Fixed-time signals operate on pre-defined schedules that cannot change with real-time changes in vehicle flow, leading to excessive waiting times and fuel wastage. Traditional actuated systems, while a little responsive, rely on threshold-based mechanisms that cannot cope with sudden changes in flow or any other unpredictable events. The introduction of IoT and AI opens up a great opportunity to bring a paradigm shift in traffic signal control. IoT can deploy intelligent sensing and communication devices, such as RSUs and OBUs, that are able to collect and transfer data in real time. Meanwhile, DRL enables a self-learning decision system to optimize control policies through continuous interaction with the environment. Further, incorporating GNNs into this framework strengthens it by capturing the spatial correlations between multiple intersections and thus supporting coordinated decision-

making instead of isolated optimization. This research aims to design, simulate, and evaluate an IoT-AI-based adaptive traffic signal control framework, integrating DRL and GNN algorithms. The proposed system would optimize the timing dynamically based on live sensor input and inter-intersection communication. Further, this framework is tested in the SUMO simulation environment to show better efficiency, reduced congestion, and environmental benefits that could be achieved for future smart mobility systems.

II. LITERATURE REVIEW

Most of the research efforts have focused on intelligent and adaptive approaches for traffic signal control, drawing on machine learning, multi-agent systems, and IoT technologies. Some early works in this direction used reinforcement learning in combination with multi-agent techniques: Abdoos et al. [1] propose a multi-agent Q-learning approach for traffic light control in a non-stationary environment. In their model, each intersection acts autonomously by means of a local adaptation to changing traffic flows. Although this system provided better responsiveness than fixed-time controllers, convergence remains slow and there is no coordination between agents. Based on this, Li et al. [2] came up with a deep reinforcement learning (DRL) method for traffic signal timing optimization, representing one of the most significant steps toward applying deep neural networks to dynamic control problems. Their approach utilized input features related to traffic state, such as vehicle density and queue length, in a DQN, which achieved better adaptability with a lower average delay compared to other conventional methods. Recent years have indeed seen an increased possibility of intelligent transportation systems because of emerging IoT technologies. AI-Absi et al. [3] proposed an adaptive traffic management system based on the IoT, which incorporates sensor data from infrastructure and vehicles for real-time signal optimization. The cloud-based data processing in their system can handle dynamic conditions, with significant improvement in throughput and response time. However, this introduced latency due to the use of centralized cloud servers, which motivated later research on distributed and edge computing methods. Liang et al. [4] surveyed the recent developments in V2V and V2I communications within the IoT ecosystem. They have pointed out that real-time communications between vehicles and traffic control units are the backbone of modern intelligent transportation frameworks. Their work shows that IoT may minimize latency and improve coordination, providing an essential technological basis for data-driven signal control. Wei et al.

[5] presented a comprehensive survey on intelligent traffic signal control using DRL. They classified existing algorithms into value-based, policy- gradient, and actor-critic categories, summarizing their strengths and limitations. Their analysis revealed that most DRL-based systems achieved local optimization but struggled to coordinate multiple intersections efficiently. Zhao et al. [6] explored computational intelligence techniques in urban traffic control, including fuzzy logic, evolutionary algorithms, and neural networks. Their review demonstrated that hybrid approaches combining rule-based heuristics with learning-based models achieved robust performance across diverse traffic conditions. This work established an early foundation for multi-agent collaboration, which later evolved into graph-based and reinforcement-learning-based paradigms. Chu et al. [7] extended DRL applications to large-scale, multi-intersection networks through a multi-agent deep reinforcement learning (MADRL) framework. Each intersection was modelled as an independent agent capable of local learning, with limited information exchange among neighbours. Their experiments on simulated traffic networks showed that cooperative agents can reduce overall travel time and mitigate congestion propagation across the network. Liu et al. [8] extended this idea with the introduction of Graph Attention Networks in a DRL framework for adaptive traffic signal control. Their model embedded spatial dependencies between intersections; hence, every agent could make a decision by considering both local and neighbouring states of traffic. The study achieved better performance in reducing queue lengths and waiting times at high levels of traffic load, indicating the strength of graph structures combined with reinforcement learning. At a theoretical level, Sutton and Barto [9] provided the foundational principles of reinforcement learning, detailing the mathematical framework underlying policy optimization and value estimation. Their seminal work serves as the theoretical backbone for most modern DRL-based traffic signal control systems. Finally, Ma et al. [10] presented a comprehensive review of GCNs applied to traffic flow forecasting. Their review highlighted the capability of GCNs to capture spatiotemporal dependencies in complex transportation networks. In fact, the integration of GCNs with DRL represents a natural evolution toward graph-based adaptive control, as explored in recent hybrid models. Collectively, these studies establish a strong foundation for IoT- and AI-driven adaptive traffic signal control. The literature clearly shows a progress from early reinforcement-learning-based approaches [1], [2] to IoT-enabled frameworks [3], [4] and finally to hybrid DRL-GNN systems [7], [8], [10] that exploit the spatial and temporal correlations. However, most existing implementations still rely on centralized computation, which limits scalability and responsiveness. The research proposed here builds on this foundation by incorporating DRL and GNN models within a distributed, edge-enabled IoT framework to provide real-time, coordinated, and scalable signal control for smart-city traffic environments.

III. PROPOSED METHODOLOGY

The architecture of the proposed IoT and AI-driven adaptive traffic control system consists of several interrelated layers, which include sensing, communication, intelligence, and application.

A. System Overview

The framework incorporates IoT sensors for monitoring the state of traffic, DRL agents for adapting to learned experience, and GNNs for collaboration between intersections. Each intersection is modelled as an independent agent communicating to other intersections through a network graph structure:

Fig. 1 depicts the system architecture overview. The lower layer represents IoT devices that consist of inductive loop detectors, cameras, and OBUs, collecting measured data such as vehicle speed, occupancy, and queue length. RSUs act as communication hubs to connect vehicles and the control system. The middle layer houses the AI decision module, where the DRL and GNN algorithms process data and decide the actions of signals. The upper layer contains edge computing nodes to perform local computation and coordinate with the cloud for data analytics and long-term policy updates.

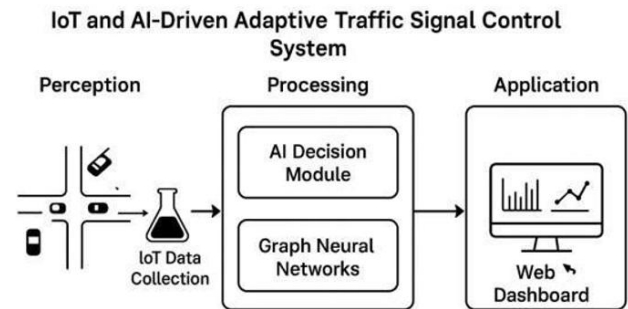


Fig. 1. System architecture of the proposed IoT and AI-driven adaptive traffic signal control framework showing IoT data acquisition, AI decision module, and edge-cloud coordination.

B. IoT-Enabled Data Acquisition

IoT sensors constitute the perception layer. Each intersection is deployed with RSUs, which receive real-time data from OBUs installed in vehicles. Some key parameters are continuously measured: the number of vehicles queuing, queue length, average speed, and elapsed signal time. These data streams are sent to the control agent via low-latency communication protocols such as MQTT or 5G V2I.

C. Deep Reinforcement Learning Agent

In the proposed framework, every intersection is considered an environment that has the traffic signal as a DRL agent. At every state (S_t), for queue lengths, waiting times, and counts of vehicles, an action of a signal phase change is taken, A_t , to decide the maximum cumulative reward, R_t . The reward function penalizes high waiting times and incentivizes improvements in throughput.

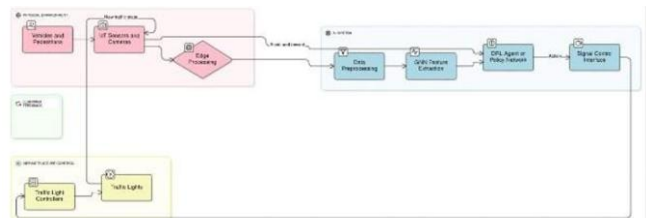


Fig. 2. Deep Reinforcement Learning (DRL) workflow illustrating interaction between environment, agent, and reward feedback loop

Three DRL algorithms were implemented: Deep Q-Network (DQN): Learns optimal state-action mappings using neural approximations of Q-values Proximal Policy Optimization (PPO): Uses clipped surrogate objectives for stable gradient updates Asynchronous Advantage Actor-

Critic (A3C): Accelerates learning through parallel training across multiple intersections

D. Graph Neural Network Coordination

A GNN models the spatial relationship between intersections. The city traffic map is represented as a graph: $G = (V, E)$, where V corresponds to an intersection and E represents connecting roads. Every node or intersection aggregates local features and exchanges information with other neighboring nodes. This coordination enables the model to capture dependencies-such as spill-back effects-and make collective decisions.

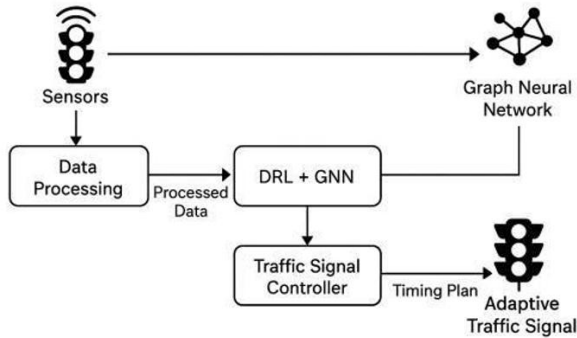


Fig. 3. Representation of multi-intersection network as a graph, showing message-passing layers between connected intersections

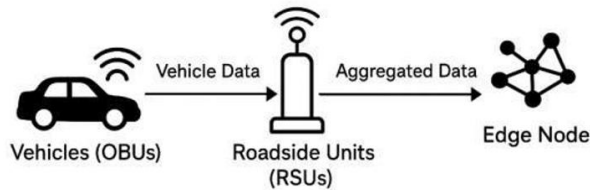


Fig. 3.1. Vehicle-to-Infrastructure (V2I) communication model illustrating real-time data exchange between On-Board Units (OBUs) and Roadside Units (RSUs) within the IoT-enabled traffic control framework.

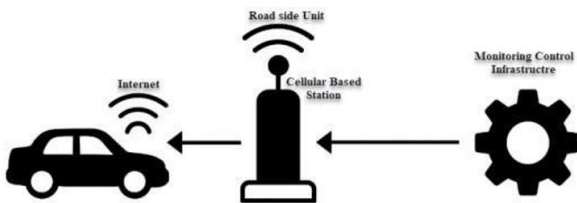


Fig. 3.2. Vehicle-to-Infrastructure (V2I) communication model illustrating data exchange between On-Board Units (OBUs) and Roadside Units (RSUs) within the IoT-enabled traffic control framework.

E. Edge Computing Integration

Edge computing is also much needed for low system latency. Without cloud-based centralized computation, each RSU conducts local inferences using the trained models so that real-time execution of decisions can be done, while periodic synchronization with the cloud will refine long-term policies.

IV. RESULTS AND DISCUSSION

A. Simulation Environment

The performance of the proposed framework was validated using SUMO (Simulation of Urban Mobility), a microscopic traffic simulation tool widely adopted in transportation research. A 4×4 grid network of intersections was modeled

to represent typical urban traffic. Each intersection was controlled by a DRL-GNN agent.

The simulation environment was developed in Python using the TraCI API, which allows real-time interaction between the SUMO simulator and external algorithms. IoT data were emulated through synthetic sensor inputs representing RSUs and OBUs.

- Hardware and Software Setup:
- Processor: Intel Core i7, 16 GB RAM
- OS: Ubuntu 22.04 LTS
- Python 3.11
- Libraries: TensorFlow, PyTorch, NetworkX, Matplotlib
- SUMO version 1.19

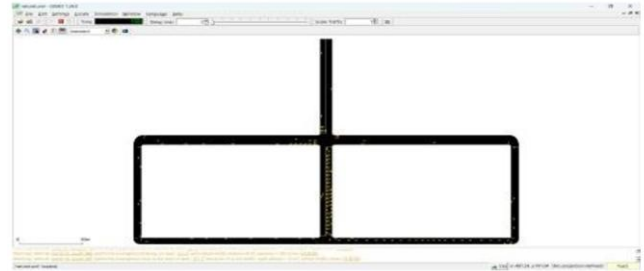


Fig. 4. SUMO simulation layout used for experimentation showing four connected intersections with vehicle inflow lanes and traffic detectors.

B. Experimental Parameters

The experiments covered multiple traffic densities: low, medium, and high inflow rates. Each intersection had four phases corresponding to north-south, east-west, and turning movements. Input states included vehicle counts, queue lengths, and waiting times per lane. The simulation ran for several thousand time steps, during which DRL models continuously updated policies based on real-time feedback. The GNN model ensured that adjacent intersections coordinated green phases to prevent queue spill-backs.

The reward function was structured to encourage shorter waiting times and higher throughput, ensuring fairness across all directions. Data were logged at each step for post-processing and visualization.

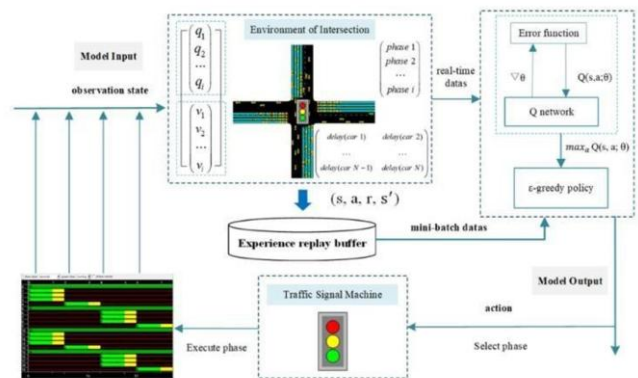


Fig. 5. Reinforcement learning training cycle showing interaction among data input, policy update, and environment response.

C. Implementation Process

- The training process consisted of the following steps:
- Initialize simulation and model parameters.
- Collect sensor data from RSUs and OBUs.
- Compute the state vector and feed it to the DRL agent.
- Execute the selected action in SUMO via TraCI.

- Update model weights based on received reward. This iterative loop continued until convergence. The trained model was then deployed for testing under new traffic conditions to evaluate generalization performance.

D. Result Analysis

Results obtained from SUMO show that the proposed DRL-GNN system adapts itself with changing traffic conditions. The adaptive control mechanism balanced signal timings across multiple intersections, leading to smoother flows and reduced waiting queues. Compared with conventional fixed-time and actuated controllers, the proposed system consistently yielded lower average delays and higher throughput in various testing scenarios. The coordination of GNN eliminated isolated optimization and prevented secondary congestion at downstream intersections. According to the visualization dashboards, there was a significantly reduced red-light waiting time for high-density scenarios. Vehicles had more continuous movement; stops per trip were fewer. Edge computing reduced command latency, thus enabling decisions in near real-time. The experimental results validate that combining IoT-based sensing, DRL decision intelligence, and GNN spatial reasoning forms a robust and scalable approach for urban traffic management.

V. CONCLUSION

IoT, DRL, and GNN technologies have been integrated for a paradigm shift in traffic signal control. While traditional centralized systems are characterized by latency and low scalability, the proposed hybrid framework distributes intelligence at each intersection. The reinforcement learning component guarantees continuous adaptation; the GNN provides coordination, which is a must for multi-intersection networks. This work affirms that local edge processing reduces computational overheads at the central servers and allows for fault tolerance. Real-world deployment presents several challenges in constrained hardware, unreliable sensor data, and interference in the network. These constraints call for further optimizations of communication protocols and security measures. Federated Learning: Intersections train models collaboratively without necessarily sharing the raw data for privacy and speedier adaptation. Multi-Model Data Fusion: Richer state representation through the introduction of real camera feeds and environmental sensors. Hybrid Cloud-Edge Architecture, Combine Centralized Long-Term Learning with Distributed Local Decision Making. Autonomous Vehicle Coordination: This framework can be extended for handling V2V interactions in a cooperative driving environment. The entire system will meet global smart-city goals of efficiency, sustainability, reduced

environmental impact, and increased productivity; its modular and scalable architecture allows easy adaptation to most city infrastructures. This research proposed an IoT-and-AI-driven adaptive traffic control framework that combined Deep Reinforcement Learning and Graph Neural Networks for intelligent decision-making. The system utilizes IoT sensors to collect real-time data, conducts local processing through edge computing, and leverages DRL for phase optimization and GNN for spatial coordination. Results from simulation in SUMO confirm the framework's capability of reducing delay, improving throughput, and enhancing sustainability. The proposed system builds the foundation for smart urban traffic management in the next generation by distributing intelligence across intersections and ensuring minimum latency. The future development will be based on large-scale field testing, integration with smart infrastructure, and adaptive learning in mixed traffic conditions.

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